# $h$-adic topology and profinite linear lambda terms

Daniel Rogozin

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## 1 Profinite Lambda terms and related concepts

One starts with the observation that if we have a work over a finite alphabet $\Sigma$:

$$W = a_{w_1} \dots a_{w_n} \in \Sigma^*$$

then one can represent $W$ as a well-typed term in the simply typed lambda calculus:

$$a_{w_1} : q \to q, \dots, a_{w_n} : q \to q \vdash W : q \to q$$

where $q$ is a base type that we informally interpret as a type of states, so we have a closed lambda term

$$\lambda a_{w_1}^{q \to q} \dots \lambda a_{w_n}^{q \to q} \lambda x^q . a_{w_k}(\dots(a_{w_1}\ q)\dots) : (q \to q) \dots (q \to q) \to q \to q$$

encoding a word $w$ via the instance of the Church encoding. Further, we will call the type of such terms $\mathrm{Church}_\Sigma$. Further, we have the monoid isomorphism:

$$\Sigma^* \cong \{M \text{ is a lambda term } | M : \mathrm{Church}_\Sigma \}/\beta\eta$$

In other words, words over a finite alphabet can be viewed as lambda terms of a particular type factorised module the $\beta\eta$-equivalence. Generally, there is amount literature studying both syntactic and semantic aspects of the Church encoding of the finite words.

The claim made in [vGMM23] connects the finitary interpretation of STLC in the category of finite sets **FinSet** with deterministic finite automata as follows. Take any finite set $Q$, so for any type $A$, we define a set $[\![A]\!]_Q$ inductively defined as

- $[\![p]\!]_Q = Q$
- $[\![A \to B]\!]_Q = [\![A]\!]_Q \Rightarrow [\![B]\!]_Q$

so every term $x_1 : A_1, \ldots, x_n : A_n \vdash M : A$ transports to the function between finite sets $[\![M]\!] : [\![A_1]\!]_Q \times \ldots \times [\![A_n]\!]_Q \to [\![A]\!]_Q$. On closed terms, the interpretation induces a function $[\![.]\!]_Q : \Lambda_{\beta\eta}(A) \to [\![A]\!]_Q$ mapping every closed term $M$ of type $A$ to its interpretation $[\![M]\!]_Q$. Further, a word $W = a_{w_1} \ldots a_{w_k} \in \Sigma^*$ is mapped to the following morphism in **FinSet**: $[\![W]\!]_Q : (Q \Rightarrow Q) \times \ldots \times (Q \Rightarrow Q) \to (Q \Rightarrow Q)$, i.e.,

$$[\![W]\!]_Q : f_{w_1}, \ldots, f_{w_k} \mapsto f_{w_k} \circ \ldots \circ f_{w_1}$$

Recall that a *deterministic finite state automaton* on a finite alphabet $\Sigma = \{a_1, \ldots, a_n\}$ is a structure $\mathcal{A} = (Q, \delta, q_0, Acc)$, where $Q$ is a finite set of states, $\delta : \Sigma \times Q \to Q$ is a transition function, $q_0 \in Q$ is the initial state and $Acc \subseteq Q$ is the set of accepting states. The transition function $\delta$ induces a family of functions $\delta_{a_i}(q_k) = \delta(a_i, q_k) : Q \to Q$ for each $a_i \in \Sigma$ and $q_k \in Q$. $\delta_{a_i}(q_k) = q_{k'}$ means that $\mathcal{A}$ transitions from $q_k$ to $q'_k$ encountering $a_i$. Now one can apply the interpretation $[\![W]\!]_Q$ to the functions $\delta_{a_{w_i}}$ for $1 \leq i \leq k$, so we obtain

$$\delta_w : [\![W]\!]_Q(\delta_{a_{w_1}}, \ldots, \delta_{a_{w_k}}) : Q \to Q$$

Therefore, given a deterministic finite automaton $\mathcal{A} = (Q, \delta, q_0, Acc)$, the tuple $(Q, \delta, q_0)$ induces the evaluation function

$$\mathrm{eval}_{\delta, q_0} : [\![\mathrm{Church}_\Sigma]\!]_Q \to Q$$

transporting every functional $F \in [\![\mathrm{Church}_\Sigma]\!]_Q$ to the state $F(\delta_{a_{w_1}}, \ldots, \delta_{a_{w_k}})$ in $Q$, therefore we have

$$\Sigma^* \cong \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma) \to [\![\mathrm{Church}_\Sigma]\!]_Q \xrightarrow{\mathrm{eval}_{\delta, q_0}} Q$$

so we associate every finite word $w \in \Sigma^*$ with the final state $q_f = \delta_w(q_0)$. The inverse image $Acc \subseteq Q$ under this composite is called a *regular language* $L_\mathcal{A}$, the set of finite words recognisable by $\mathcal{A}$.

Regular languages $L_\mathcal{A}$ form a Boolean algebra $\mathrm{Reg}_Q(\mathrm{Church}_\Sigma)$ of regular languages of lambda terms of type $\mathrm{Church}_\Sigma$ recognised by $Q$. The algebra can be defined as the image of the Boolean algebra homomorphism $[\![.]\!]_Q^{-1} : 2^{[\![\mathrm{Church}_\Sigma]\!]} \to 2^{\Sigma^*}$ obtained by applying the contravariant powerset functor $2^{(.)} : \mathbf{FinSet}^{op} \to \mathbf{BA}$ to the semantic bracket $[\![.]\!]_Q$. The Boolean algebra $\mathrm{Reg}_Q(A)$ of regular languages of lambda terms of any higher order type $A$ recognisable by a finite set $Q$ is defined as the image of the Boolean algebra homomorphism $[\![.]\!]_Q^{-1} : 2^{[\![A]\!]_Q} \to 2^{\Lambda_{\beta\eta}(A)}$. That is, a set $L$ of lambda terms of type $A$ is recognisable by the finite set $Q$ exactly when it has the form $[\![.]\!]_Q^{-1}(\mathrm{Acc}) = \{M \in \Lambda_{\beta\eta}(A) \mid [\![M]\!]_Q \in \mathrm{Acc}\}$ for some choice $\mathrm{Acc} \subseteq [\![A]\!]_Q$ of accepting elements. Thus, the collection $\mathrm{Reg}(A) \subseteq 2^{\Lambda_{\beta\eta}(A)}$ of lambda terms of type $A$ as follows

$$\mathrm{Reg}(A) = \bigcup \{\mathrm{Reg}_Q(A) \mid Q \in \mathbf{FinSet}\}.$$

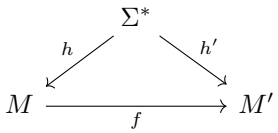
The monoid $\widehat{\Sigma^*}$ is the free profinite monoid generated by $\Sigma$. $\widehat{\Sigma^*}$ can be constructed as the limit, computed in the category of monoids **Mon**, of the projective system of finite monoid homomorphisms

$$(\Sigma^*/\phi \to \Sigma^*/\phi')_{\phi \subseteq \phi'}$$

where $\phi$ and $\phi'$ are finite index congruences on the free monoid $\Sigma^*$ such that $\phi \subseteq \phi'$. Every finite index congruence $\phi$, in turn, induces a surjective homomorphism $h : \Sigma^* \to M$ for the quotient monoid $M = \Sigma^*/\phi$. The surjectivity condition on $h$ can be relaxed to show that $\widehat{\Sigma^*}$ of profinite words is the codirected limit of the composite functor

$$\Sigma^*/\mathbf{FinMon} \xrightarrow{\pi} \mathbf{FinMon} \hookrightarrow \mathbf{Mon}$$

$\Sigma^*/\mathbf{FinMon}$ is the slice category whose objects $(M, h)$ are pairs consisting of a finite monoid $M$ and a homomorphism $h : \Sigma^* \to M$ and whose morphisms $(M, h) \to (M', h')$ are homomorphisms $f : M \to M'$ such that the following triangle commutes:




The projection functor $\pi : \Sigma^*/\mathbf{FinMon} \to \mathbf{FinMon}$ transports $(M, h)$ to its underlying monoid $M$. One obtains $\widehat{\Sigma^*}$ as the limit of a codirected diagram of monoid homomorphisms $(M \to M')_{(M,h)\to(M',h')}$ extending the diagram formed from the family $(\Sigma^*/\phi \to \Sigma^*/\phi')_{\phi\subseteq\phi'}$ to all homomorphisms $h : \Sigma^* \to M$ to a finite monoid $M$.

Recall that every homomorphism $h : \Sigma^* \to M$ induces a deterministic finite automaton with $Q := M$, $\delta(a, q) = q \cdot_M a$ for every $q \in M$ and $a \in \Sigma$. So every homomorphism $h : \Sigma^* \to M$ (assume $M = \{q_1, \ldots, q_m\}$) induces a decomposition of $\Sigma^*$ into $m$ components $L_{q_i} = h^{-1}(q_i)$ for $1 \leq i \leq m$ where each $L_{q_i}$ is a regular language. Let $\mathrm{Reg}_{(M,h)}(\Sigma)$ denote the Boolean algebra of languages of the form $L_q = h^{-1}(q)$ for $q \in M$. We have a functor:

$$\mathrm{Reg}_{(.)}(\Sigma) : (\Sigma^*/\mathbf{FinMon})^{op} \to \mathbf{BA}$$

The Boolean algebra $\mathrm{Reg}_{(M,h)}(\Sigma)$ coincides with $2^h$, the image of $h : \Sigma^* \to M$ applied to the contravariant powerset functor. Also, the following directed diagram in $\mathbf{BA}$ associated to $\mathrm{Reg}_{(.)}(\Sigma)$:

$$(\mathrm{Reg}_{(M',h')}(\Sigma) \to \mathrm{Reg}_{(M,h)}(\Sigma))_{(M,h)\to(M',h')}$$

can be obtained by applying the contravariant powerset functor to the above trianlge and $(M \to M')_{(M,h)\to(M',h')}$.

## 1.1 The Explicit Description of Profinite Topology

This description is based on [Pin10, C.X.2]. Let $\Sigma$ be a finite alphabet and let $M$ be a finite monoid, given $u, v \in \Sigma^*$, a monoid homomorphism $f : \Sigma^* \to M$ *separates* $u$ and $v$ if $f(u) \neq f(v)$. A monoid $M$ separates $u$ and $v$ if there is a homomorphism $\Sigma^* \to M$ separating $u$ and $v$. The important fact is that every $u$ and $v$ are separated by some monoid $M$.

A metric on $\Sigma^*$ is defined as follows: two words are closed if they are separated by the large monoid. Take $u, v \in \Sigma^*$, put:

- $r(u, v) = \min\{|M| \mid M \text{ separates } u \text{ and } v\}$,
- $d(u, v) = 2^{-r(u,v)}$.

Therefore, $d(u, v) < 2^n$ iff $u$ and $v$ cannot be separated by a monoid smaller than $n$ and $d(u, v) < \varepsilon$ iff $u$ and $v$ cannot be separated by a monoid of size $n$, where $n$ is the unique natural number such that $2^{-(n+1)} < \varepsilon \leq 2^{-n}$. $d$ is a metric on $\Sigma^*$ such that $(\Sigma^*, d)$ is an ultrametric space such that, for each $u, v, w \in \Sigma^*$:

- $d(uw, vw) \leq d(uv)$,
- $d(wu, wv) \leq d(uv)$.

As a topological space, $(\Sigma^*, d)$ is discrete, but open balls are of more interest. For any $n \geq 2$, consider the congruence relation $\sim_n$ on $\Sigma^*$ defined as:

$$u \sim_n v \Leftrightarrow \forall M\ (|M| \leq n \to (\forall f : \Sigma^* \twoheadrightarrow M) \to f(u) = f(v))$$

Clearly $\sim_n$ is a congruence of finite index as they are finitely many surjections $\Sigma^* \twoheadrightarrow M$ for a fixed finite $M$. Thus $d(u, v) < 2^{-n}$ iff they cannot be separated by a monoid of size smaller than $n$.

**Proposition 1.1.** Let $\Sigma$ be a finite alphabet,

1. The concatenation function $(u, v) \mapsto uv$ is uniformly continuous,
2. Every open ball of $(\Sigma^*, d)$ is a recognisable language and every recognisable language is a finite union of open balls.
3. $(\Sigma^*, d)$ is totally bounded: for every $n > 0$, $\Sigma^*$ is covered by a finite number of open balls of radius strictly smaller than $2^{-n}$.

**Proposition 1.2.** Let $\Sigma_1$ and $\Sigma_2$ be finite alphabets, then $f : \Sigma_1 \to \Sigma_2$ is uniformly continuous iff for every recognisable language of $\Sigma_2$ $L$, $f^{-1}(L)$ is also recognisable.

Let $\widehat{\Sigma^*}$ denote the completion of $\Sigma^*$, whose points are called profinite words of $\Sigma$.

**Proposition 1.3.** Let $M$ be a finite monoid and let $f : \Sigma^* \to M$ be uniformly continuous, then $f$ uniquely extends to $\widehat{f} : \widehat{\Sigma^*} \to M$.

The following is the universal property of the free profinite monoid:

**Proposition 1.4.** Let $\Sigma$ be a finite alphabet, then if $f : \Sigma \to M$ is a function to a finite monoid $M$, there exists a unique uniformly continuous monoid morphism $\widehat{f} : \widehat{\Sigma^*} \to M$ such that for each $a \in \Sigma$, $\widehat{f}(a) = f(a)$. $\widehat{f}$ is onto iff $f(\Sigma)$ generates $M$.

**Proposition 1.5.** Let $\Sigma_1$ and $\Sigma_2$ be finite alphabets, then every uniformly continuous morphism $f : \Sigma_1^* \to \Sigma_2^*$ uniquely extends to uniformly continuous $\widehat{f} : \widehat{\Sigma_1^*} \to \widehat{\Sigma_2^*}$.

The metric on $\widehat{\Sigma^*}$ is defined similarly. Let $M$ be a finite monoid, then a morphism $f : \Sigma^* \twoheadrightarrow M$ separates profinite words $u, v \in \widehat{\Sigma^*}$ if $\widehat{f}(u) = \widehat{f}(v)$. For $u, v \in u, v \in \widehat{\Sigma^*}$, we set:

- $r(u, v) = \min\{|M| \mid M \text{ separates } u \text{ and } v\}$,
- $d(u, v) = 2^{-r(u,v)}$.

**Proposition 1.6.** The profinite topology on $\widehat{\Sigma^*}$ is the least topology making every continuous every morphism $\Sigma^*$ onto a finite discrete monoid.

## 1.2 $\omega$-terms

Let $x \in \widehat{\Sigma^*}$ be a profinite word, the profinite word $x^\omega$ is defined as

$$x^\omega = \lim_{n\to\infty} x^{n!}$$

The existence of $x^\omega$ follows from the following proposition:

**Proposition 1.7.** For each $x \in \widehat{\Sigma^*}$, the sequence $(x^{n!})_{n\in\mathbb{N}}$ is a Cauchy sequence converging to an idempotent element of $\widehat{\Sigma^*}$.

Note that $x^\omega$ is a choice of notation, but it is not an infinite word. One needs to intepret $\omega$ as the exponent of a finite semigroup. One can compute the image $x^\omega$ under a morphism to a finite monoid.

Let $M$ be a finite monoid with exponent $\omega$, let $f : \Sigma^* \to M$ and let $s = f(x)$ for $x \in \Sigma^*$. Then the sequence $s^{n!}$ converges to the idempotent $s^\omega$ of the subsemingroup of $M$ generated by $s$. Therefore, one obtains

$$\hat{f}(x^\omega) = f(x)^\omega$$

Also, $(x^\omega)^\omega = x^\omega$ and $x^{\omega\omega} = x^\omega$. Besides, we have:

$$x^{\omega+1} = \lim_{n\to\infty} x^{n!+1}$$
$$x^{\omega-1} = \lim_{n\to\infty} x^{n!-1}$$

So we have by the definition:

$$xx^\omega = x^{\omega+1} = x^\omega x$$
$$xx^{\omega-1} = x^\omega = x^{\omega-1}x$$

One also gets $f(x^{\omega+1}) = s^{\omega+1}$, though the interpretation of $x^{\omega-1}$ is slightly more complicated. The minimal ideal of the subsemigroup generated by $s$ is a group $G$ with identity $s^\omega$. But $s^{\omega-1}$ might not be in $G$. However, in $M$ one has $\lim_{n\to\infty} s^{n!-1} = s^{2\omega-1}$, and, thus, $f(x^{\omega-1}) = s^{2\omega-1}$. Besides, $s^{2\omega-1}$ is the inverse of $s^{\omega+1}$ in $G$.

The $\omega$-terms on $\Sigma$ is the smallest submonoid of $\widehat{\Sigma^*}$ containing $\Sigma^*$ and closed under operations $x \mapsto x^\omega$, $x \mapsto x^{\omega+1}$ and $x \mapsto x^{\omega-1}$.

## 1.3 Profinite Lambda Terms

One moves from profinite words to profinite lambda terms in the following way. For any finite set $Q$, one has the interpretation of any type $A$ as the finite set $[\![A]\!]_Q$. Given another finite set $Q'$ with a relation $R \subseteq Q \times Q'$, one can relate different interpretations of $A$ by constructing a relation $[\![A]\!]_R \subseteq [\![A]\!]_Q \times [\![A]\!]_{Q'}$. Such relations are called *logical relations*. Lambda terms are parametric: that is, if $M$ has type $A$ and $R \subseteq Q \times Q'$, then $([\![M]\!]_Q, [\![M]\!]_{Q'}) \in [\![A]\!]_R$.

Now let us discuss how one can define the exponential of relations. Let $S \subseteq P \times P'$ and $R \subseteq Q \times Q'$ be relations on finite sets, the exponential $S \Rightarrow R \subseteq (P \Rightarrow Q) \times (P' \Rightarrow Q')$ is given as:

$$S \Rightarrow R := \{(g,h) \mid \forall x \in P \ \forall y \in P' \ (xSy \to g(x)Rh(y))\}$$

Therefore, for any relation $R \subseteq Q \times Q'$ between finite sets $Q$ and $Q'$, one constructs a relation $[\![A]\!]_R \subseteq [\![A]\!]_Q \times [\![A]\!]_{Q'}$ defined by induction on $A$ as follows:

- $[\![p]\!] = R$,

- $[\![A \Rightarrow B]\!]_R = [\![A]\!]_R \Rightarrow [\![B]\!]_R$

The *fundamental lemma of logical relations* states that for any $M \in \Lambda_{\beta\eta}(A)$ for any $A$ and for any $R \subseteq Q \times Q'$, the interpretations of $M$ are related as follows:

$$([\![M]\!]_Q, [\![M]\!]_{Q'}) \in [\![A]\!]_R$$

Further, we will be using partial surjections since they are stable under taking relational exponentials. Also, if $f : Q \twoheadrightarrow Q'$ is a partial surjection, so $[\![A]\!]_f \subseteq [\![A]\!]_Q \times [\![A]\!]_{Q'}$ for any $A$. Besides, we have the inclusion of Boolean algebras $\mathrm{Reg}_{Q'}(A) \subseteq \mathrm{Reg}_Q(A)$, which is proved using the fundamental lemma of logical relations.

**Theorem 1.1.** The diagram of Boolean algebras $\mathrm{Reg}_{(.)}(A) : \mathbf{FinPSurj}^{op} \to \mathbf{BA}$:

$$(\mathrm{Reg}_{Q'}(A) \hookrightarrow \mathrm{Reg}_Q(A))_{f:Q \to Q' \in \mathbf{FinPSurj}}$$

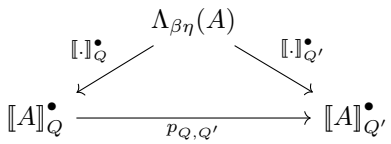
is directed, and its colimit in **BA** coincides with the Boolean algebra $\mathrm{Reg}(A)$ of regular languages for a higher-order type $A$.

Also, for any simple type $A$, the set of regular languages $\mathrm{Reg}(A)$ is closed under finite intersections.

Now let us discuss how the space of profinite lambda terms is constructed. The idea is that for any simple type $A$ as parametric families of semantic elements, so they form a Stone space dual to the Boolean algebra $\mathrm{Reg}(A)$.

**Proposition 1.8.** Let $Q, Q'$ be finite sets,

1. The set of atoms $\mathrm{At}_Q(A)$ of the finite Boolean algebra $\mathrm{Reg}_Q(A)$ is in bijection with the set $[\![A]\!]^\bullet_Q$ of definable elements of simple type $A$, given by the function $q \mapsto [\![.]\!]_Q^{-1}(\{q\}) : \mathrm{At}_Q(A) \to [\![A]\!]^\bullet_Q$.
2. For partial surjection $f : Q \twoheadrightarrow Q'$ and for any simple type $A$, the set $[\![A]\!]^\bullet_Q \subseteq [\![A]\!]_Q$ of definable elements is contained in the domain of $[\![A]\!]_f$, and the restriction of $[\![A]\!]_f$ to $[\![A]\!]^\bullet_Q$ is the unique function $p_{Q,Q'}$ such that the following triangle commutes:




Note that if $f, g : Q \to Q'$ are partial surjections, then, while the semantic interpretations $[\![A]\!]_f, [\![A]\!]_g : [\![A]\!]_Q \to [\![A]\!]'_Q$ are different, their restrictions to $[\![A]\!]^\bullet_Q$ are both equal to $p_{Q,Q'}$.

One defines profinite lambda terms of simple type $A$ as follows.

**Definition 1.1.** Let $A$ be a simple type, the set of *profinite lambda terms* is defined as the limit $\widehat{\Lambda}_{\beta\eta}(A)$ in **Set** of the diagram:

$$([\![A]\!]^\bullet_f : [\![A]\!]^\bullet_Q \twoheadrightarrow [\![A]\!]^\bullet_{Q'})_{f:Q \twoheadrightarrow Q' \in \mathbf{FinPSurj}}$$

In particular, by the definition of a limit in **Set**, it means that a profinite lambda term in $\widehat{\Lambda_{\beta\eta}(A)}$ is a family $\theta$ of definable terms $\theta_Q \in [\![A]\!]_Q$ for every finite set $Q$ such that for every partial surjection

$f : Q \twoheadrightarrow Q'$ one has $[\![A]\!]^\bullet_f(\theta_Q) = \theta_{Q'}$. By Proposition 1.8, the above condition is equivalent to the condition that for every term $M : A$ for simple $A$ and any finite sets $Q$ and $Q'$:

$$\theta_Q = [\![M]\!]_Q \wedge |Q'| \leq |Q| \Rightarrow \theta_{Q'} = [\![M]\!]_{Q'}$$

$\widehat{\Lambda_{\beta\eta}(A)}$ is equipped with a natural topology making it a Stone dual of the Boolean algebra $\mathrm{Reg}(A)$. One can define this topology as inherited from the inclusion

$$\widehat{\Lambda_{\beta\eta}(A)} \hookrightarrow \prod_Q [\![A]\!]^\bullet_Q$$

into the product space $\prod_Q [\![A]\!]^\bullet_Q$ where each $[\![A]\!]^\bullet_Q$ is considered a space with the discrete topology. Concretely, for any finite set $Q$ and $q \in [\![A]\!]^\bullet_Q$, let $U_{Q,q}$ denote the set of profinite lambda terms taking value at $Q$:

$$U_{Q,q} := \{\theta \in \widehat{\Lambda_{\beta\eta}(A)} \mid \theta_Q = q\}$$

Now we take the collection $U_{Q,q}$ as the basis on $\widehat{\Lambda_{\beta\eta}(A)}$ for any $Q$ and $q$.

The following is similar to [Geh16, Section 4.2]:

**Proposition 1.9.** The space $\widehat{\Lambda_{\beta\eta}(A)}$ is a Stone dual to the Boolean algebra $\mathrm{Reg}(A)$.

## 1.4 The cartesian closed category of profinite lambda terms

Let $A$ be simple type, a *parametric family* is a family of points $\theta_Q \in [\![A]\!]_Q$ for a finite set $Q$, such that for any relation $R \subseteq Q \times Q'$ one has $\theta_Q [\![A]\!]_R \theta_{Q'}$. Every parametric family is preserved under partial surjections. Thus, any parametric family whose components are definable elements is a profinite term.

**Theorem 1.2.** Profinite terms $\theta \in \widehat{\Lambda}_{\beta\eta}(A)$ can be equivalently characterised as parametric families of definable terms $\theta_Q \in [\![A]\!]^\bullet_Q$.

Profinite lambda terms are assembled in a cartesian closed category by using the construction from [JM18].

Let $\mathcal{C}$ be a cartesian closed category quipped with a functor $\mathcal{P} : \mathcal{C} \to \mathbf{Set}$ with cartesian products preserved as follows. The canonical functions

$$(\mathcal{P}\pi_1, \mathcal{P}\pi_2) : \mathcal{P}(A \times B) \to \mathcal{P}A \times \mathcal{P}B$$
$$!_{\mathcal{P}\top} : \top \to \mathcal{P}\top$$

have the following inverses:

$$\mathsf{m}_{A,B} : \mathcal{P}A \times \mathcal{P}B \to \mathcal{P}(A \times B)$$
$$\mathsf{m}_\top : \mathcal{P}\top \to \top$$

Let us define the category $\mathcal{C}[\mathcal{P}]$ whose objects are the objects of $\mathcal{C}$ and whose hom-sets are defined as

$$\mathcal{C}[\mathcal{P}](A, B) = \mathcal{P}(B^A)$$

**Proposition 1.10.** The category $\mathcal{C}[\mathcal{P}]$ is cartesian closed and comes equipped with the identity-on-object (cartesian closed) functor $\mathbf{idonobj}_{\mathcal{C}[\mathcal{P}]} : \mathcal{C} \to \mathcal{C}[\mathcal{P}]$ strictly preserving exponentials and cartesian products.

Let us recall how the cartesian closed category **Lam** of simply typed lambda terms is generated by the terminal category.

**Definition 1.2.** The category **Lam** has objects as simple types and hom-sets $\mathbf{Lam}(A, B)$ are defined as $\Lambda_{\beta\eta}(A \Rightarrow B)$ for simple types $A$ and $B$.

Let us consider the functor $\mathcal{P} : \mathbf{Lam} \to \mathbf{Set}$ defined as $\mathcal{P}(A) = \widehat{\Lambda_{\beta\eta}}(A)$. $\mathcal{P}$ sends every term $M : A \Rightarrow B$ to the set-theoretic function sending a profinite term $\theta$ of type $A$ on the family $(\llbracket M \rrbracket_Q(\theta_Q))$ that can be shown to be profinite term of type $B$ via the fundamental lemma of logical relations. One has

$$\mathcal{P}(A \times B) \cong \mathcal{P}(A) \times \mathcal{P}(B)$$
$$\mathcal{P}\top \cong \top$$

Therefore, we obtain a cartesian closed category $\mathbf{ProLam} := \mathbf{Lam}(\mathcal{P})$ whose hom-sets are $\mathbf{ProLam}(A, B) := \widehat{\Lambda_{\beta\eta}}(A \Rightarrow B)$.

The functor $\mathcal{P}$ is valued in **Set**, but it also can be value in any cartesian category $\mathcal{S}$ as long as $\mathcal{P}$ isomorphically preserves cartesian products, we will get the cartesian category $\mathcal{C}[\mathcal{P}]$ enriched over $\mathcal{S}$. In particular, the functor $\mathcal{P} : \mathbf{Lam} \to \mathbf{Set}$ factors through the category of Stone spaces as follows:

$$\mathbf{Lam} \xrightarrow{\widehat{\Lambda}_{\beta\eta}(.)} \mathbf{Stone} \xrightarrow{\text{forget}} \mathbf{Set}$$

The category **ProLam** is equipped with a cartesian closed idenity-on-functor functor $\mathbf{idonobj} : \mathbf{Lam} \to \mathbf{ProLam}$ derived from the fact that **Lam** is a free cartesian closed category.

**Proposition 1.11.** The functor $\mathbf{idonobj} : \mathbf{Lam} \to \mathbf{ProLam}$ is faithful.

*Proof.* The category **ProLam** can be obtained as the limit of a codirected diagram of cartesian closed category described as follows. Let $Q$ be a finite set and let $A$ be a simple type, consider the following equivalence relation:

$$\sim_Q^A \subseteq \Lambda_{\beta\eta}(A) \times \Lambda_{\beta\eta}(A)$$

on the set of terms of type $A$ modulo $\beta\eta$-equivalence defined as

$$M \sim_Q^A N \leftrightarrow \llbracket M \rrbracket_Q = \llbracket N \rrbracket_Q$$

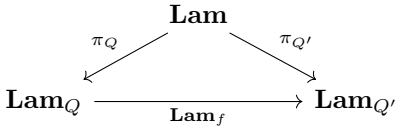
Fix a finite set $Q$, the family $\sim_Q$ is parametrised by simple types defines a congruence relation on the category **Lam**:

$$f \sim_Q^{A \Rightarrow B} f' \wedge g \sim_Q^{B \Rightarrow C} g' \Rightarrow g \circ f \sim_Q^{A \Rightarrow C} g' \circ f'$$

Therefore we obtain the category $\mathbf{Lam}_Q := \mathbf{Lam}/Q$ with morphisms defined as:

$$\mathbf{Lam}_Q(A, B) = \Lambda_{\beta\eta}(A \Rightarrow B)/\sim_Q^{A \Rightarrow B}$$

First of all, one can observe that for any finite set $Q$, $\mathbf{Lam}_Q$ is cartesian closed and equipped with cartesian closed identity-on-functor $\pi_Q : \mathbf{Lam} \to \mathbf{Lam}_Q$. Besides, every partial surjection $f : Q \twoheadrightarrow Q'$ in **FinPSurj** induces a cartesian closed identity-on-functor $\mathbf{Lam}_f : \mathbf{Lam}_Q \to \mathbf{Lam}_{Q'}$ such that the following triangle commutes:



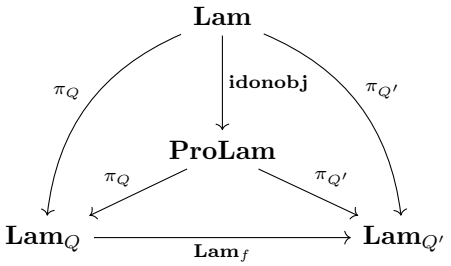
Further, one observes that the category **ProLam** is the codirected limit of the diagram of cartesian closed categories $\mathbf{Lam}_Q$ indexed by finite sets and partial surjections between them. The projection functor $\pi_Q : \mathbf{ProLam} \to \mathbf{Lam}_Q$ is defined by transporting every morphism $\theta : A \to B$ defined as a family $\theta$ of definable elements in

$$\mathbf{ProLam}(A, B) = \widehat{\Lambda}_{\beta\eta}(A \Rightarrow B)$$

to the corresponding $\theta_Q$ in

$$\mathbf{Lam}_Q(A \Rightarrow B) = \widehat{\Lambda}_{\beta\eta}(A \Rightarrow B)/_Q^{A \Rightarrow B} \cong [\![A \Rightarrow B]\!]_Q^\bullet$$

so the canonical functor makes the following diagram commute:




Further, one concludes that **idonobj** is faithful as follows. By Statman's finite completeness theorem [Sta82], for every pair of terms $M, N : A \to B$, then either $M \equiv_{\beta\eta} N$ or there is a finite set $Q$ such that the interpretations $[\![M]\!]_Q = \pi_Q(M)$ and $[\![N]\!]_Q = \pi_Q(N)$ are different. In this case $\mathbf{idonobj}(M)$ and $\mathbf{idonobj}(N)$ are different as well, and therefore **idonobj** is faithful. □

Further, let us recall some facts from the higher-order language theory.

**Proposition 1.12.** Let $\Sigma$ be a finite alphabet, then the Boolean algebras $\mathrm{Reg}(\mathrm{Church}_\Sigma)$ and $\mathrm{Reg}(\Sigma)$ given by the Church encoding.

*Proof.* By the Church encoding, there is a one-to-one correspondence between $L \subseteq \Sigma^*$ and $\mathcal{L} \subseteq \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma)$. One needs to show that $L \subseteq \Sigma^*$ is regular iff the associated element $\mathcal{L} \subseteq \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma)$ belongs to $\mathrm{Reg}(\mathrm{Church}_\Sigma)$.

Let $L$ be a language recognised by a deterministic finite automaton $\mathcal{A} = (Q, \delta, q_0, \mathrm{Acc})$. The associated set $\mathcal{L} \subseteq \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma)$ is defined by the inverse image of the semantic bracket $[\![.]\!]_Q : \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma) \to [\![\mathrm{Church}_\Sigma]\!]_Q$ of the set of functionals in $[\![\mathrm{Church}_\Sigma]\!]_Q$ defined in the following way:

$$\mathrm{eval}^{-1}_{\delta,q_0}(\mathrm{Acc}) = \{F \in [\![\mathrm{Church}_\Sigma]\!]_Q \mid F(\delta_{a_1}, \ldots, \delta_{a_n})(q_0) \in \mathrm{Acc}\}$$

Therefore $\mathcal{L}$ is an element of $\mathrm{Reg}_Q(\mathrm{Church}_\Sigma)$, and therefore, an element of $\mathrm{Reg}(\mathrm{Church}_\Sigma)$.

Conversely, by definition of $\mathrm{Reg}(\mathrm{Church}_\Sigma)$, it suffices to show that for every finite set $Q$, a subset $\mathcal{L} \in \mathrm{Reg}_Q(\mathrm{Church}_\Sigma)$ corresponds to some regular language $L \subseteq \Sigma^*$. We can suppose that $\mathcal{L}$ has the following form

$$\mathcal{L} = [\![.]\!]_Q^{-1}(\{F\}) = \{M \in \Lambda_{\beta\eta}(\mathrm{Church}_\Sigma) \mid [\![M]\!]_Q = F\}$$

where $F$ is a functional in $[\![\mathrm{Church}_\Sigma]\!]_Q$. The corresponding set $L \subseteq \Sigma^*$ is the finite intersection of regular languages recognised by the deterministic finite automata of the form $\mathcal{A} = (Q, \delta, q_0, \{q_f\})$ where the unique finite state $q_f$ is equal to $q_f = F(\delta_{a_1}, \ldots, \delta_{a_n})(q_0)$, therefore the set $L$ is a regular language itself being the finite intersection of regular languages. □

By Stone duality and the above theorem, one also has that for every finite alphabet $\Sigma$, there is a homeomorphism $\widehat{\Lambda}_{\beta\eta}(\mathrm{Church}_\Sigma) \cong \widehat{\Sigma^*}$ between the space $\widehat{\Lambda}_{\beta\eta}(\mathrm{Church}_\Sigma)$ of profinite lambda terms $\mathrm{Church}_\Sigma$ and the space of profinite words $\widehat{\Sigma^*}$.

This is how the concept of an $\omega$-word transferred to lambda terms.

**Proposition 1.13.** Let $A$ be a simple type, then there exists a term $\Omega_A : (A \Rightarrow A) \Rightarrow A \Rightarrow A$ which, for any profinite term $M \in \widehat{\Lambda}_{\beta\eta}(A \Rightarrow A)$ satisfies the idempotency equation:

$$(\Omega_A M) \circ (\Omega_A M) = \Omega_A M$$

# 2 $h$-adic topology

The following summary is based on [Kas12, Chapter XVI].

Consider the complex algebra $K = \mathbb{C}[[h]]$ of complex formal series in one variable $h$. Every element $f \in \mathcal{K}$ has the form

$$f = \sum_{n=0}^{\infty} a_n h^n$$

where $a_n \in \mathbb{C}$ for each $n \in \mathbb{N}$. Every polynomial $f \in \mathbb{C}[h]$ is clearly a formal series. $K$ form a commutative ring with operations given as

- $f + g = \sum\limits_{n=0}^{\infty} a_n h^n + \sum\limits_{n=0}^{\infty} b_n h^n = \sum\limits_{n=0}^{\infty} (a_n + b_n) h^n$
- $f \cdot g = (\sum\limits_{n=0}^{\infty} a_n h^n) \cdot (\sum\limits_{n=0}^{\infty} b_n h^n) = \sum\limits_{n=0}^{\infty} (\sum\limits_{n=p+q} a_p b_q) h^n$

A formal series $f$ is invertible iff its every coefficient is non-zero.

For any $n > 0$, consider the algebra $K_n = \mathbb{C}[h]/(h^n)$ of truncated polynomials obtained as the quotient of the ring of complex polynomials generated by the ideal $(h^n)$. There is a morphism

$$\pi_n : K \to K_n$$

sending each formal series $\sum\limits_{n=0}^{\infty} a_n h^n$ to the class of $\sum\limits_{k=0}^{n-1} a_k h^k$ modulo $(h^n)$. $\pi_n$ is onto and its kernel is the ideal $h^n K$ generated by $h^n$ in the ring of formal series. Therefore, we have the following isomorphism:

$$\mathbb{C}[[h]]/(h^n) \cong \mathbb{C}[h]/(h^n)$$

For $n > 0$, there is a surjective map of algebras $p_n : K_n \to K_{n-1}$ generated by the inclusion of ideals $(h^n) \subset (h^{n-1})$. So we have the inverse system of algebras $(K_n, p_n)_{n>0}$ and its inverse limit $\varprojlim\limits_n K_n$. One has $p_n \circ \pi_n = \pi_{n-1}$ for each $n$. Also, there is a unique morphism of algebras from $K$ to $\varprojlim\limits_n K_n$ whose composition with the projection of the inverse limit is equal to $\pi_n$. Besides, the map $\pi : \mathbb{C}[[h]] \twoheadrightarrow \varprojlim\limits_n \mathbb{C}[[h]]/(h^n)$ is the isomorphism, which is partly the instance of Birkhoff's subdirect representation theorem, [SB81, Theorem 8.6].

Let us equip $K$ with the *h-adic topology*. The family of of subsets $\pi_n^{-1}(U_n)$ for $n > 0$ and $U_n \subseteq K_n$ form an open basis on $K$. $\{0\}$ is a family of open neighbourhoods in the discrete set $K_n$, the family $\pi_n^{-1}(0)$ is a family of open neighbourhoods of 0 in $K$ for the $h$-adic topology. There is

also the structure of a metric space on $K$ given as follows. Given a formal series $f = \sum_{n=0}^{\infty} a_n h^n$, let $o(f)$ denote the unique natural number such that $a_{o(f)} \neq 0$ and $a_k = 0$ for each $k < o(f)$. For $f = 0$, set $o(f) = +\infty$ and extend $(\mathbb{N}, <)$ by adding the top element $+\infty$. It is easy to see that:

$$(h^n) = \{f \in K \mid o(f) > n - 1\}$$

and therefore

$$\bigcap_{n>0} (h^n) = \{0\}$$

We also have for any series $f, g \in K$:

$$o(f + g) \geq \min(o(f), o(g))$$

Now define $|.| : K \to \mathbb{R}_{\geq 0}$ as

$$|f| = 2^{-o(f)}$$

Then if we put $d(f, g) = |f - g|$, then $(\mathbb{C}[[h]], d)$ becomes an ultrametric space.

Let $M$ be a left $K$-module, consider the family $(h^n M)_{n>0}$ of submodules of $M$ with the canonical $K$-linear projections:

$$p_n : M_n = M/h^n M \to M_{n-1} = M/h^{n-1} M$$

They form an inverse system of $K$-modules, so we can consider the inverse limit

$$\widetilde{M} = \varprojlim_n M_n$$

having a natural structure of a left $K$-module. $\widetilde{M}$ has a natural topology as well, the inverse limit topology, for which the family of submodules $(h^n \widetilde{M})_n$ form an open basis. The projections $i_n : M \to M_n$ induce a unique $K$-linear map $i : M \to \widetilde{M}$ such that $\pi_n \circ i = i_n$ for each $n$. The kernel of $i$ is given as

$$\mathrm{Ker}(i) = \bigcap_{n>0} h^n M$$

A $K$-module $M$ is *separated* if $\mathrm{Ker}(i) = \{0\}$. It is *complete* if $i$ is injective.

For any module $K$, the module $M/\mathrm{Ker}(i)$ is separated and the completion $\widetilde{M}$ of $M$ is complete.

Generally, any separated complete $K$-module $M$ is equipped with the $h$-adic topology induced by the inverse limit topology on $\widetilde{M}$ via $M \cong \widetilde{M}$.

Now let us describe the following class of separated complete $K$-modules. Let $V$ be a vector space over $\mathbb{C}$, let $V[[h]]$ denote the set of all formal series in variable $h$

$$f = \sum_{n=0}^{\infty} a_n h^n$$

where $a$'s range over elements of $V$. There is a left $K$-module on $V[[h]]$, and such modules are called *topologically free modules*. One can observe that any topologically free module is separated and complete. Also, if $\{e_i\}_{i \in I}$ is a basis of $V$, then the $K$-submodule generated by $\{e_i\}_{i \in I}$ is dense in $V$ for the $h$-adic topology. Besides, let $N$ be a complete separated $K$-module, there is a natural isomorphism in the category of $K$-modules and $K$-linear maps:

$$\mathrm{Hom}_K(V[[h]], N) \cong \mathrm{Hom}_K(V, N)$$

Further, a $K$-module $M$ is topologically free iff it is complete, separated and torsion-free, that is, there is such that $mh \neq 0$ for any non-zero $m \in M$.

Let $M$ and $N$ be left modules over the algebra $K = \mathbb{C}[[h]]$. The $K$-module $M \otimes_K N$ is the quotient of the vector space $M \otimes N$ obtained by spanning all elements of the form $fm \otimes n - m \otimes fn$ for $f \in K$, $m \in M$ and $n \in N$.

**Definition 2.1.** The topological tensor product $M\widetilde{\otimes}N$ of $M$ and $N$ is the $h$-adic completion of $M \otimes_K N$:

$$M\widetilde{\otimes}N := \widetilde{M \otimes_K N} = \varprojlim_{n>0}(M \otimes_K N)/h^n(M \otimes_K N)$$

By definition, $M\widetilde{\otimes}N$ is complete.

Take $m \in M$ and $n \in N$, let $m\widetilde{\otimes}n$ denote the image of $m \otimes n$ under the following image:

$$M \otimes N \to M \otimes_K N \to M\widetilde{\otimes}N$$

The subspace of the topological tensor product spanned by all elements of this form is dense in $M \otimes N$.

**Proposition 2.1.**

1. The class left modules over the algebra $K = \mathbb{C}[[h]]$ with $\widetilde{\otimes}$ is endowed with the structure of a symmetric monoidal category
2. If $M$ and $N$ be topologically free modules, so is $M\widetilde{\otimes}N$, that is:

$$V[[h]]\widetilde{\otimes}W[[h]] = (V \otimes W)[[h]].$$

## 2.1 Topological Algebras

A *topological algebra* is a triple $(A, \mu, \eta)$ where $A$ is a module over the ring $K = \mathbb{C}[[h]]$ and $\mu : A\widetilde{\otimes}A \to A$ and $\eta : A \to K$ are $k$-linear maps such that:

- $\mu \circ (1_A\widetilde{\otimes}\mu) = \mu \circ (\mu\widetilde{\otimes}1_A)$,
- $\mu \circ (\eta\widetilde{\otimes}1_A) = 1_A = \mu \circ (1_A\widetilde{\otimes}\eta)$.

For $a, a' \in A$, $aa'$ will stand for $\mu(a\widetilde{\otimes}a')$.

Let $(A, \mu, \eta)$ be a topological algebra and let $f(h) = \sum c_n h^n$ be a formal series with complex coefficients. For $a$, the formula

$$f(ha) = \sum_{n\geq 0} c_n a^n h^n$$

defines a unique element in the inverse limit $\widetilde{A} = \varprojlim_n A/h^n A$. If $A$ is separated and complete, then it defines an element $f(ha)$ in $A \cong \widetilde{A}$. This procedure can be applied to the exponential function $e^h = \sum_{n\geq 0} \frac{h^n}{n!}$ yeilding elements of the form

$$e^{ha} = \sum_{n\geq 0} \frac{a^n h^n}{n!}$$

in any separate complete topological algebra. If $a' \in A$, then $a'$ *commutes* with $a$ whenever

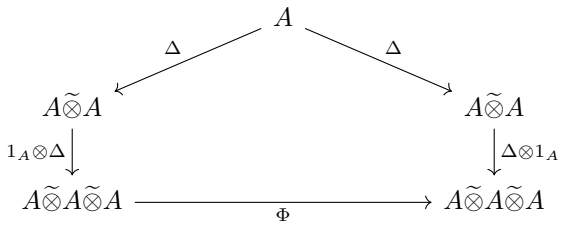
$$e^{a+a'} = e^{ha}e^{ha'}$$

$e^{ha}$ is also invertible in $A$ with $e^{-ha}$.

A morphism $f : (A, \mu, \eta) \to (A', \mu', \eta')$ of topological algebras is a $K$-linear map $f : A \to A'$ such that $f \circ \mu = \mu' \circ (f \widetilde{\otimes} f)$ and $f \circ \eta = \eta'$.

A topological quasi-bialgebra is a structure $(A, \mu, \eta, \Delta, \varepsilon, \Phi)$, where $(A, \mu, \eta)$ is a topological algebra, $\Delta : A \to A\widetilde{\otimes}A$ and $\varepsilon : A \to K$ are $K$-linear maps and $\Phi : A^{\widetilde{\otimes}3} \cong A^{\widetilde{\otimes}3}$:

- The following commutes:




- $(A, \mu, \eta)$ is a topological coalgebra,
- $(1_A\widetilde{\otimes}1_A\widetilde{\otimes}\Delta)(\Phi)(\Delta\widetilde{\otimes}1_A\widetilde{\otimes}1_A)(\Phi) = \Phi_{234}(1_A\widetilde{\otimes}\Delta\widetilde{\otimes}1_A)(\Phi)\Phi_{123}$,
- $(1_A\widetilde{\otimes}\varepsilon\widetilde{\otimes}1_A) = 1 \otimes 1$.

A topological quasi-bialgebra is a topological bialgebra whenever $\Phi = 1^{\widetilde{\otimes}3}$.

A topological quasi-bialgebra $(A, \mu, \eta, \Delta, \varepsilon, \Phi)$ is a *topological braided quasi-algebra* if there is an invertible element $R \in A\widetilde{\otimes}A$ such that

- $\Delta^{op}(a) = R\Delta(a)R^{-1}$ for any $a \in A$,
- $(1_A\widetilde{\otimes}\Delta)(R) = (\Phi_{231})^{-1}R_{13}\Phi_{213}R_{12}(\Phi_{123})^{-1}$
- $(\Delta\widetilde{\otimes}1_A)(R) = \Phi_{321}R_{13}(\Phi_{132})^{-1}R_{23}\Phi_{123}$

$R$ is called the universal $R$-matrix of $A$.

A *gauge transformation* on a topological quasi-algebra is an invertible element $F \in A\widetilde{\otimes}A$ such that

$$(1_A\widetilde{\otimes}\varepsilon_A)F = (\varepsilon_A\widetilde{\otimes}1_A)F = 1$$

**Example 2.1.** Let $A_0 = (A_0, \mu_0, \eta_0, \Delta_0, \varepsilon_0, \Phi_0, R_0)$ be a braided quasi-bialgebra over $\mathbb{C}$. One can define the following topological quasi-braided algebra:

$$A_0[[h]] = (A_0[[h]], \mu, \eta, \Delta, \varepsilon, \Phi_0, R_0)$$

on the space of formal series with coefficients in $A_0$ where $\mu, \eta, \Delta, \varepsilon$ are unique $K$-linear maps such that

$$\forall f \in K \;\; \eta(f) = f1 = f\eta_0 1$$

and for all $a, a' \in A$

- $\mu(a\widetilde{\otimes}a') = \mu_0(a \otimes a')$,

- $\Delta(a) = \Delta_0(a)$,
- $\varepsilon(a) = \varepsilon_0(a)$.

$A_0[[h]]$ is called the trivial topological braided quasi-algebra associated to $A_0$.

**Example 2.2.** Let $A = (A, \mu, \eta, \Delta, \varepsilon, \Phi, R)$ be a topological braided quasi-algebra. We have $(A\widetilde{\otimes}A)/h(A\widetilde{\otimes}A) \cong A/hA \otimes A/hA$, so the $K$-linear maps $\mu, \eta, \Delta, \varepsilon$ induce $\mathbb{C}$-linear maps:

- $\overline{\mu} : A/hA \otimes A/hA \to A/hA$,
- $\overline{\Delta} : A/hA \to A/hA \otimes A/hA$,
- $\overline{\eta} : \mathbb{C} \to A/hA$,
- $\overline{\varepsilon} : A/hA \to \mathbb{C}$.

Define $\overline{\Phi}$ as the class of $\Phi$ modulo $(A/hA)^{\otimes 3}$ and define $\overline{R}$ as the class of $R$ modulo $(A/hA)^{\otimes 3}$, then $\overline{A} = (\overline{A}, \overline{\mu}, \overline{\eta}, \overline{\Delta}, \overline{\varepsilon}, \overline{\Phi}, \overline{R})$ is a braided quasi-algebra.

A *topological $A$-module* $M$ over a topological algebra $(A, \mu, \eta)$ is a left $K$-module with a $K$-linear map $\mu_M : A\widetilde{\otimes}M \to M$ such that

$$\begin{array}{ccc} A \otimes A \otimes M & \xrightarrow{\mu\widetilde{\otimes}1_M} & A\widetilde{\otimes}M \\ {\scriptstyle 1_A\widetilde{\otimes}\mu_M}\downarrow & & \downarrow{\scriptstyle \mu_M} \\ A \otimes M & \xrightarrow[\mu_M]{} & M \end{array} \qquad \begin{array}{ccc} M & \xrightarrow{\eta\widetilde{\otimes}1_M} & A \otimes M \\ & \searrow & \downarrow{\scriptstyle \mu_M} \\ & & M \end{array}$$

One writes $\mu_M(a\widetilde{\otimes}m) := am$ for $a \in A$ and $m \in M$. Let $M$ and $N$ be topological $A$-modules, then their tensor product $M\widetilde{\otimes}N$ is a topological $A\widetilde{\otimes}A$-module. If $A$ has a comultiplication $\Delta : A \to A\widetilde{\otimes}A$. One can pull back the $A\widetilde{\otimes}A$-module structure on $M\widetilde{\otimes}N$ to a topological $A$-module structure given as follows, for $a \in A$, $m \in M$ and $n \in N$:

$$a(m\widetilde{\otimes}n) = \Delta(a)(m\widetilde{\otimes}n)$$

Let $A$ be a topological braided algebra with universal $R$-matrix $R$, then for any topological $A$-module $M$ the $k$-linear automorphism $c^R_{M,M}$ is defined by

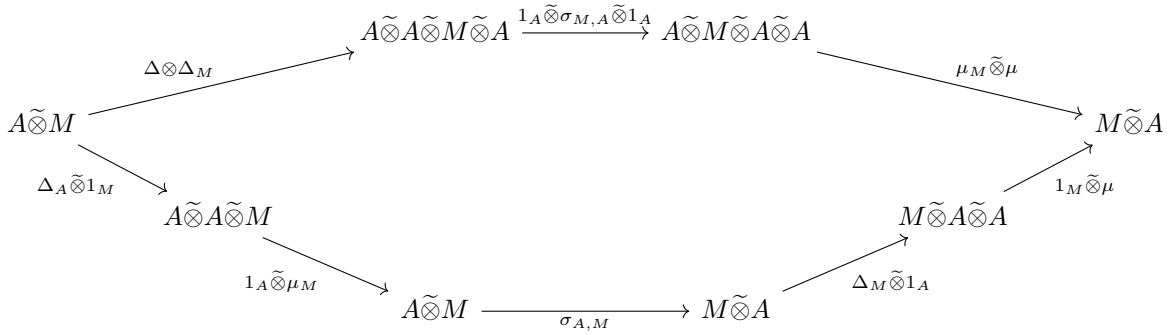
$$c^R_{M,M}(m_1\widetilde{\otimes}m_2) = (R(m_1\widetilde{\otimes}m_2))_{21}$$

is a solution of the Yang-Baxter equation in $M^{\widetilde{\otimes}3}$.

The universal $R$-matrix of a topological braided quasi-bialgebra gives rise to a representation of of the braid group $\mathcal{B}_n$ on the topological $A$-module $M^{\widetilde{\otimes}n}$ for each $n > 1$ and a topological $A$-module $M$.

A topological $A$-module $M$ over a topological bialgebra $A = (A, \mu, \eta, \Delta, \varepsilon)$ equipped with a left action $\mu_M : A\widetilde{\otimes}M \to M$ is a *topological crossed $A$-module* if there is a $k$-linear map $\Delta_M : M \to M\widetilde{\otimes}A$ such that:

$$\begin{array}{ccc} M & \xrightarrow{\Delta_A} & M\widetilde{\otimes}A \\ {\scriptstyle \Delta_A}\downarrow & & \downarrow{\scriptstyle \Delta_M\widetilde{\otimes}A} \\ M\widetilde{\otimes}A & \xrightarrow[1_M\widetilde{\otimes}\Delta]{} & M\widetilde{\otimes}A\widetilde{\otimes}A \end{array} \qquad \begin{array}{ccc} M & \xrightarrow{\Delta_A} & M\widetilde{\otimes}A \\ & \searrow & \downarrow{\scriptstyle 1_M\widetilde{\otimes}\varepsilon} \\ & & M \end{array}$$




## 2.2 Quantum Enveloping Algebras

**Definition 2.2.** A *complex Lie algebra* is a vector space $\mathfrak{g}$ over $\mathbb{C}$ with the operation $[.,.] : \mathfrak{g}\times\mathfrak{g}\to\mathfrak{g}$ satisfying the following axioms for all $x, y, z \in \mathfrak{g}$ and $a, b \in \mathbb{C}$:

1. $[x, x] = 0$,
2. $[x, [y, z]] + [y, [x, z]] + [z, [x, y]] = 0$,
3. $[ax + by, z] = a[x, z] + b[y, z]$,
4. $[z, ax + by] = a[z, x] + b[z, y]$.

Any complex Lie algebra $\mathfrak{g}$ has a universal enveloping algebra $U(\mathfrak{g})$ equipped with the structure of bialgebra as follows for all $x \in U(\mathfrak{g})$:

- $\varepsilon(x) = 0$,
- $\Delta(x) = 1 \otimes x + x \otimes 1$

$U(\mathfrak{g})$ also has a trivial braided quasi-bialgebra structure with $\Phi = 1 \otimes 1 \otimes 1$ and $R = 1 \otimes 1$. By the Poincaré–Birkhoff–Witt theorem, every Lie algebra $\mathfrak{g}$ embeds into $U(\mathfrak{g})$.

**Definition 2.3.** A *quantum enveloping algebra* (QUE) for the Lie algebra $\mathfrak{g}$ is a topological braided quasi-bialgebra $(A, \mu, \eta, \Delta, \varepsilon, \Phi, R)$ such that $A$ is a topologically free module, and the induced braided quasi-algebra $\overline{A} = (\overline{A}, \overline{\mu}, \overline{\eta}, \overline{\Delta}, \overline{\varepsilon}, \overline{\Phi}, \overline{R})$ as in Example 2.2 coincides with $U(\mathfrak{g})$ and $\eta$ coincides with $\overline{\eta}$.

First of all, a QUE is topologically free, so $A = (A/hA)[[h]]$ as a left $K$-module. Therefore $A/hA = U(\mathfrak{g})$. So $A = U(\mathfrak{g})[[h]]$ as a $K$-module. So we derive from Proposition 2.1:

$$\forall n > 0 \;\; A^{\widetilde{n}} = (U(\mathfrak{g})^{\otimes n})[[h]]$$

Further, maps $\mu$, $\eta$, $\Delta$ and $\varepsilon$ are determined by their restrictions to $U(\mathfrak{g})\otimes U(\mathfrak{g})$, $\mathbb{C}$, $U(\mathfrak{g})$ and $U(\mathfrak{g})$ respectively. For $a, a' \in U(\mathfrak{g})$ we have

$$\mu(a \otimes a') = \sum_{n\geq 0} \mu_n(a \otimes a')h^n$$

where $\{\mu_n : U(\mathfrak{g})\otimes U(\mathfrak{g}) \to U(\mathfrak{g}) \,|\, n \geq 0\}$ is a family of linear maps such that $\mu_0$ is the multiplication in the enveloping algebra. Similarly we have:

$$\forall a \in U(\mathfrak{g})\ \ \Delta(a) = \sum_{n\geq 0} \Delta_n(a)h^n$$
$$\forall a \in U(\mathfrak{g})\ \ \sum_{n\geq 0} \varepsilon_n(a)h^n$$
$$\forall f \in \mathbb{C}[[h]]\ \ \eta(f) = f1$$

By Proposition 2.1, we also write:

$$\Phi = \sum_{n\geq 0} \Phi_n h^n$$
$$R = \sum_{n\geq 0} R_n h^n$$

By definition, a quantum enveloping algebra $A$ is associated to a Lie algebra $\mathfrak{g}$. $\mathfrak{g}$ is recovered from $A$ as

$$\mathfrak{g} = \{x \in A/hA \mid \Delta_0(a) = 1 \otimes a + a \otimes 1\}$$

# 3 Duality for Topological $[[h]]$-modules and topological Hopf algebras

This section is loosely based on [Gav02].

Fix a field $k$, so $k[[h]]$ is by default a topological ring, that is, it is equipped with the $h$-adic topology. Let $X$ be any $k[[h]]$-module, then we set

$$X_0 := X/hX = k \otimes_{k[[h]]} X,$$

a $k$-module via scalar restriction

$$k[[h]] \to k[[h]]/hk[[h]] \cong k$$

that we call *the specialisation of $X$ at $h = 0$*, or, *the semiclassical limit of $X$*. Let

$$X \xrightarrow{h\to 0} \overline{Y}$$

We have discussed that if $X$ is complete, torsion-less and separated, then there is a natural isomorphism of $k$-modules $X \cong X_0[[h]]$.

Let we set ${}^F X := k((h)) \otimes_{k[[h]]} X$, a vector space over $k((h))$ without any topology.

Let $X$ be a topological $k[[h]]$-module, we let:

- $X^* := \mathrm{Hom}_{k[[h]]}(X, k[[h]])$, the *full dual* of $X$,
- $X^\star := \{f \in X^* \mid f \text{ is continuous}\}$, the *topological dual* of $X$.

There are two monoidal categories of topological $k$-modules:

- $\mathcal{T}_{\widehat{\otimes}}$, which is modelled on the monoidal category of discrete topological $k$-vector spaces $k$-vector spaces.
- $\mathcal{P}_{\widetilde{\otimes}}$, which is modelled on the monoidal category of linearly compact topological $k$-vector spaces.

The monoidal category $\mathcal{T}_{\widehat{\otimes}}$ consists of all topologically free modules (that is, modules isomorphic to $V[[h]]$ for some $k$-vector space $V$) and $k[[h]]$-linear maps. The tensor product $T_1 \widehat{\otimes} T_2$ is the separated $h$-adic completion of the algebraic tensor product $T_1 \otimes_{k[[h]]} T_2$ for $T_1, T_2 \in \mathcal{T}_{\widehat{\otimes}}$.

The monoidal category $\mathcal{P}_{\widetilde{\otimes}}$ is the closure under isomorphism of the class of topological modules of type $k[[h]]^E$ (the cartesian product indexed by $E$ with the Tikhonov product topology) for some set $E$. These modules are complete w.r.t the weak topology. They are isomorphic to the projective limit of their finite free submodule (with the $h$-adic topology). Morphisms in $\mathcal{P}_{\widetilde{\otimes}}$ are $k[[h]]$-linear continuous maps. The tensor product $P_1 \widetilde{\otimes} P_2$ for $P_1, P_2 \in \mathcal{P}_{\widetilde{\otimes}}$ is the completion of $P_1 \otimes_{k[[h]]} P_2$ w.r.t. the weak topology: thus, if $P_i \cong k[[h]]^{E_i}$ for $i = 1, 2$, then one has $P_1 \widetilde{\otimes} P_2 \cong k[[h]]^{E_1 \times E_2}$.

Note that objects in both $\mathcal{T}_{\widehat{\otimes}}$ and $\mathcal{P}_{\widetilde{\otimes}}$ are complete w.r.t. the $h$-adic topology.

Further, $\mathcal{HA}_{\widehat{\otimes}}$ is the subcategory of $\mathcal{T}_{\widehat{\otimes}}$ whose objects are Hopf algebras in $\mathcal{T}_{\widehat{\otimes}}$ and whose morphisms are Hopf algebra morphisms in $\mathcal{T}_{\widehat{\otimes}}$. Similarly, $\mathcal{HA}_{\widetilde{\otimes}}$ is the subcategory of Hopf algebras from $\mathcal{P}_{\widetilde{\otimes}}$ whose morphisms are Hopf algebra morphisms in $\mathcal{P}_{\widetilde{\otimes}}$. $\mathcal{HA}_{\widetilde{\otimes}}^{\omega - I}$ is the full subcategory of $\mathcal{HA}_{\widetilde{\otimes}}$ consisting of all $H \in \mathcal{HA}_{\widetilde{\otimes}}$ whose weak topology coincides with the $I_H$-adic topology where

$$I_H := hH + \mathrm{Ker}(\varepsilon) = \varepsilon^{-1}(hk[[h]])$$

Note that objects in $\mathcal{HA}_{\widehat{\otimes}}$ and $\mathcal{P}_{\widetilde{\otimes}}$ are *topological Hopf algebras*, in particular, the comultiplication in those algebras is given as:

$$\Delta(x) = \sum_{(x)} x_{(1)} \otimes x_{(2)}$$

# 4 On Dialogue Categories and their internal language

**Definition 4.1.** A dialogue category is a monoidal category $(\mathcal{C}, \otimes, \mathbb{1})$ equipped with an object $\perp$ along with a family of isomorphisms

$$\varphi_{A,B} : \mathcal{C}(A \otimes B, \perp) \cong \mathcal{C}(B, A \multimap \perp)$$

natural in $x$ and $y$; along with a family of isomorphisms

$$\psi_{A,B} : \mathcal{C}(A \otimes B, \perp) \cong \mathcal{C}(A, \perp \circ\!\!- B)$$

natural in $A$ and $B$ as well.

## 4.1 Term Assignment for Tensorial Logic

Figure 1: Term Calculus for Tensorial Logic

$$\frac{}{\vdash \mathbf{1} : \mathbb{1}}\ \mathbb{1}\mathbf{I} \qquad \frac{}{x : A \vdash x : A}\ \mathbf{ax} \qquad \frac{\Gamma \vdash M : \mathbb{1} \qquad \Delta \vdash N : B}{\Gamma, \Delta \vdash \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ N : B}\ \mathbb{1}\mathbf{E}$$

$$\frac{x : A, \Gamma \vdash M : \bot}{\Gamma \vdash \nabla_l x.M : A \multimap \bot}\ \multimap \bot_{\mathbf{I}} \qquad \frac{\Delta \vdash N : A \qquad \Gamma \vdash M : A \multimap \bot}{\Gamma, \Delta \vdash N \triangleleft M : \bot}\ \multimap \mathbf{E}$$

$$\frac{\Gamma, x : A \vdash M : \bot}{\Gamma \vdash \nabla_r x.M : \bot \circ\!\!- A}\ \bot \circ\!\!-_{\mathbf{I}} \qquad \frac{\Gamma \vdash M : A \circ\!\!- \bot \qquad \Delta \vdash N : A}{\Gamma, \Delta \vdash M \triangleright N : \bot}\ \multimap \mathbf{E}$$

$$\frac{\Gamma \vdash M : A \qquad \Delta \vdash N : B}{\Gamma, \Delta \vdash M \otimes N : A \otimes B}\ \otimes\mathbf{I} \qquad \frac{\Gamma \vdash M : A \otimes B \qquad \Delta, x : A, y : B \vdash N : C}{\Delta, \Gamma \vdash \mathbf{let}\ x \otimes y\ =\ M\ \mathbf{in}\ N : C}\ \otimes\mathbf{E}$$

**Definition 4.2.** Reduction rules for tensorial logic:

$(\multimap \bot\beta)$ $N \triangleleft (\nabla_l x.M) \to M[x := N]$,

$(\bot \circ\!\!- \beta)$ $(\nabla_r x.M) \triangleright N \to M[x := N]$,

$(\mathbb{1}\beta)$ $\mathbf{let}\ \mathbf{1} = \mathbf{1}\ \mathbf{in}\ N \to N$,

$(\otimes\beta)$ $\mathbf{let}\ x \otimes y = M \otimes N\ \mathbf{in}\ P \to P[x := M, y := N]$,

$(\multimap \bot\eta)$ $(\nabla_l x.x \triangleleft M) \to M$ if $x$ is not free in $M$,

$(\bot \circ\!\!- \eta)$ $(\nabla_r x.M \triangleright x) \to M$ if $x$ is not free in $M$,

$(\otimes\eta)$ $\mathbf{let}\ x \otimes y = M\ \mathbf{in}\ N[z := x \otimes y] \to N[z := M]$,

$(\otimes_{\mathbf{nat}})$ $P[w := \mathbf{let}\ x \otimes y = M\ \mathbf{in}\ N] \to \mathbf{let}\ x \otimes y = M\ \mathbf{in}\ P[w := N]$,

$(\mathbb{1}_{\mathbf{nat}})$ $P[w := \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ N] \to \mathbf{let}\ \mathbf{1} = M\ \mathbf{in}\ P[w := N]$.

In non-commutative tensorial logic, contexts in antecedents are lists, but in the formulation with antecedents as multisets, then we can just unify left and right negations.

## 4.2 Exponential Negation

Questions:

- Do we need resource modalities? If so, then have a look at [MT10], [MTT18] and [MT07]. Exponential negation? Note sure what's more essential for the profinite setting.
- Should tensorial terms be symmetric (as well as dialogue categories)? At least, we'll have less boilerplate in the reduction rules.
- Do we need sums? Sums are essential for the resource-sensitive CPS as they model non-deterministic choice.

- When we model the profinite lambda terms, we use the category of finite sets as the starting point: intuitively, **FinSet** is the cartesian closed category of finite types. What's the analogue of **FinSet** for tensorial terms? $\mathbb{F}$-weighted automaton over Galois fields? Some references: [BR11], [MN23], [AMU18], [UACM16], [BBB$^+$12], [VZ14].
- How one can describe the constuction with the $\mathcal{P}$-functor from [JM18] for dialogue categories?

### 4.3 Parametricity for Tensorial Logic

The following subsection develops parametricity for the tensorial calculus in the fashion of [AMP25] and [ZZZ10].

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